

SHIVAM PAWAR

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Education

Manipal University Jaipur

B.Tech in Computer Science and Engineering(AI & ML)

August 2024 – Present

CGPA-8.95

Projects

Asthma Prediction System with Explainable AI | *Python, ML, SHAP, LIME* September 2025-March 2026

- Developed a machine learning model to predict asthma risk using clinical and environmental data.
- Integrated explainability techniques (SHAP and LIME) to generate interpretable predictions for better understanding.
- Achieved **98.29% accuracy** using an SVM (RBF kernel) classifier trained on a 10,000-record clinical dataset with 14 features.
- Designed a structured pipeline for data processing, model training, and evaluation.
- Generated detailed explanations using a language model to improve clarity of results.

Customer Segmentation using Advanced Unsupervised Learning | *Scikit-learn, Pandas, Matplotlib* January 2026

- Engineered an end-to-end customer segmentation pipeline leveraging multiple clustering algorithms including K-Means, Hierarchical Clustering, and DBSCAN.
- Transformed raw transactional data through extensive preprocessing, feature engineering, and outlier handling to improve data quality and model reliability.
- Optimized clustering performance using the Elbow Method and Silhouette Score, enabling data-driven selection of optimal cluster configurations.
- Uncovered meaningful customer patterns and behavioral segments through exploratory data analysis and advanced visualizations.

Healthcare Prediction using Machine Learning Models | *Python, Scikit-learn, Pandas* December 2025

- Developed and evaluated classification models including Logistic Regression and Boosting algorithms on healthcare datasets.
- Performed data preprocessing, feature selection, and handled missing values to improve model accuracy and reliability.
- Analyzed model performance using evaluation metrics and comparative analysis to identify optimal predictive models.

Publications

LLMs as Explainable Layer for ML-based Asthma Prediction Model | *Manuscript in Preparation* April 2026

- Research paper focused on interpretable machine learning for healthcare prediction.
- Manuscript currently in preparation, focusing on validation and refinement of experimental results.

Technical Skills

Languages: Python, SQL, JavaScript, HTML, CSS, C, MATLAB

Machine Learning: Supervised Learning, Unsupervised Learning, Explainable AI (SHAP, LIME), Ensemble Methods (Boosting), NLP, Data Preprocessing

Libraries/Frameworks: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Flask/FastAPI (learning)

Tools: Git, GitHub, Jupyter Notebook, VS Code

Operating Systems: Windows, Linux

Relevant Coursework

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|-----------------------------------|---|---|
| • Data Structures & Algorithms | • Database Management Systems | • Calculus & Matrices |
| • Operating Systems(OS) | • Principles of Artificial Intelligence | • Cloud Computing |
| • Object-Oriented Programming | • Statistics & Probability | • Software Engineering & Project Management |
| • Design & Analysis of Algorithms | • Regression Analysis & Forecasting | |

Certifications

Introduction to HTML, CSS, & JavaScript

IBM, Coursera ([View Certificate](#))

May 2026

Machine Learning with Python

IBM, Coursera ([View Certificate](#))

December 2025

Generative AI for Software Development

DeepLearning.AI, Coursera ([View Certificate](#))

December 2025

Python Essentials 1 & 2

Cisco Networking Academy ([View Certificate](#))

October 2025

Leadership / Extracurricular

Club Garuda

April 2025 – March 2026

Head of Finance and Registration

Manipal University Jaipur

- Spearheaded financial planning and budget strategy for large-scale technical and cultural events, ensuring efficient allocation of resources.
- Managed and optimized an 11 lakh budget for Dandiya Night 2025, successfully supporting an event attended by 5000+ participants.
- Directed end-to-end financial operations and streamlined registration processes to enhance event efficiency and user experience.
- Coordinated cross-functional teams to execute logistics for a national-level hackathon (CodeVeda), ensuring smooth and timely operations.